

# TECHNICAL REPORT

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Project:  
Customer Churn Business Analysis

### 1. Introduction

This project focuses on customer churn analysis using business datasets.

### 2. Dataset Information

Datasets used:

- customer\_churn.csv
- sales\_data.csv

### 3. Data Cleaning

- Removed duplicates
- Handled missing values
- Converted data types

### 4. Exploratory Data Analysis

Performed:

- Churn distribution analysis
- Revenue analysis
- Correlation analysis
- Customer segmentation

### 5. Statistical Analysis

Performed:

- T-test
- Confidence interval analysis
- Correlation analysis

### 6. Predictive Modeling

Logistic Regression was used for churn prediction.

### 7. Results

The predictive model successfully identified churn patterns.

### 8. Business Recommendations

- Improve customer retention
- Focus on high-risk customers
- Build retention campaigns

### 9. Conclusion

The project demonstrated a complete business analytics workflow.